DSCI441 COURSE POSTER

Used Car Price Prediction System

End-to-end Craigslist price regression pipeline with baseline comparison, nonlinear modeling, extension diagnostics, and deployment-facing web demo.

BEST BROAD-DOMAIN R²
0.8061
HistGradientBoosting (`hgb_ordinal`)

BEST BROAD-DOMAIN RMSE
$6,649
Test split (80/20, seed 441)

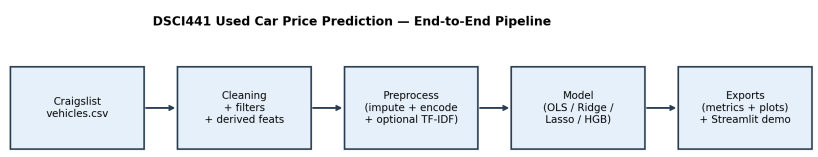
TOP-DECILE MAE
$10.1k
High-price listings show the largest absolute-dollar miss

PROBLEM DESCRIPTION / MOTIVATION

Why used-car price regression is not a single-metric task

Price prediction must handle heavy tails, missingness, and nonlinear segment interactions. A single global RMSE can hide large differences across price ranges and vehicle segments.

- Target is strongly right-skewed.
- Many categorical fields are sparse/missing.
- Error scale increases with true price (heteroskedasticity).



Consistent workflow from cleaning to model export and interactive demo.

DATASET DESCRIPTION

Craigslist listings with mixed tabular structure

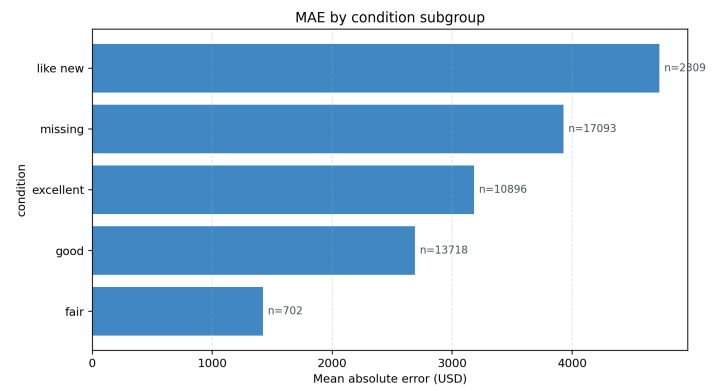
The dataset contains 426,880 raw listings with numeric, categorical, and optional text fields. Cleaning and feature derivation produce a stable training table for fair model comparison.

426,880
Raw listings

17+
Core input attributes

Heavy-tail
Price distribution shape

Missingness
High in several categorical columns



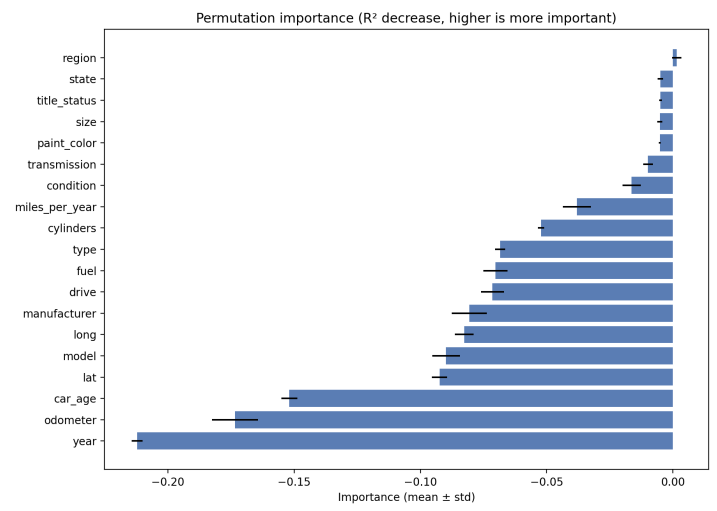
Extension diagnostics confirm subgroup-level MAE variation.

EXPERIMENTS / MODEL SELECTION

Linear baselines, text-augmented ridge, and boosted trees

Experiments include OLS, Ridge, Lasso, Ridge+TF-IDF, and HistGradientBoosting. The nonlinear model gives the strongest broad-domain fit while preserving a reproducible pipeline.

Boosted trees lead in broad-domain R².



Age/mileage and key segment attributes dominate signal contribution.

Why nonlinear wins: it captures segment interactions that linear baselines only approximate.

Why baselines stay: OLS/Ridge/Lasso still anchor interpretation and set a reproducible floor.

EVALUATION

Global fit is strong, but error behavior is segment-dependent

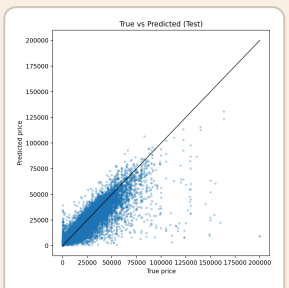
The best model achieves high broad-domain fit, yet error distribution differs by price decile and subgroup. Extension diagnostics expose these deployment-relevant patterns.

- The headline model is strong enough to summarize the broad market, but not uniform enough to justify one confidence message.
- High-price listings remain the most visible source of larger absolute-dollar miss.
- The extension therefore shifts the project from score reporting to decision-aware evaluation.

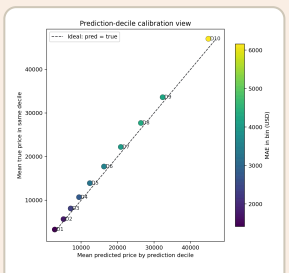
Broad-domain best: R² 0.8061, RMSE 6649, MAE 3384.

Decile drift: MAE rises from about $1.8k to about $10.1k (D1 -> D10).

MAE and mean residual shift by true-price decile.



High-price tail remains harder even with best overall model.



Decile-level predicted vs true means reveal calibration-style drift.

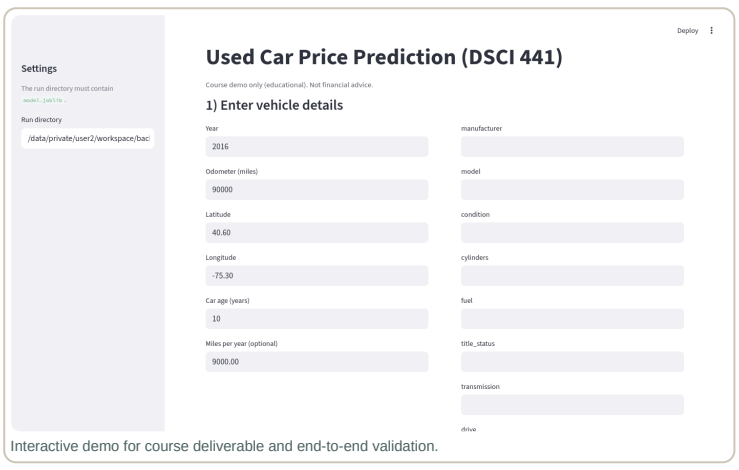
What the figures show
The model is useful overall, but it compresses some market regions and loses precision in the expensive tail.

Why that matters
A user-facing system should report more than one scalar score when uncertainty is segment-dependent.

DEPLOYMENT AND INTERPRETABILITY

From offline metrics to user-facing diagnostics

The Streamlit app provides an input -> predict interface. The extension recommends pairing predictions with subgroup/decile diagnostics so users can interpret uncertainty by market segment.
In other words, the poster is not claiming a fully production-ready valuation engine. It shows that the course project now includes the right packaging layers: reproducible training, comparative evaluation, demo exposure, and a clearer reading of failure modes.



Interactive demo for course deliverable and end-to-end validation.

Recommended production view: show global metrics and segment-aware diagnostics together.

Future direction: segment-specific models and interval predictions for risk-aware pricing.

- The GitHub Pages companion turns the same message into a static, deployable walkthrough.
- The poster, report, and website now tell the same core story across formats.

How the upgraded diagnostics should be used

The main contribution of the extension is not a new winning model; it is a better interpretation protocol for using the same predictor responsibly in a user-facing setting.

- Show the point estimate together with a segment-aware uncertainty note.
- Treat premium listings as higher absolute-dollar risk cases even with good global R².
- Use subgroup plots to flag where future segment-specific models may be worth the cost.

For demos
Report estimate, range, and listing attributes together.

For grading
Keep aggregate metrics, but add decile and subgroup behavior.

For future work
Explore intervals, segment models, and richer text signals.

What was built
Training pipeline, baseline comparison, nonlinear model, diagnostics, report, poster, and static site.

What was learned
Broad accuracy and deployment confidence are related, but not equivalent.

What remains open
Intervals, segment-specific models, and stronger geographic robustness are next logical extensions.

Why the packaging matters
The same findings now exist in report, poster, static site, and demo form, which makes the project easier to review and reuse.

Why it is still honest
The added materials improve communication and diagnosis, but they do not pretend the model is a finished commercial pricing engine.

Takeaway

This project demonstrates a complete DSCI441 workflow: reproducible training, strong nonlinear performance, and an upgraded diagnostic layer that turns aggregate metrics into deployment-relevant insights.